

Inference

Last time

$$\begin{aligned}\text{Admit}_i &\sim \text{Bernoulli}(\pi_i) \\ \log\left(\frac{\pi_i}{1 - \pi_i}\right) &= \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ &\quad \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i\end{aligned}$$

Research question: Is there a relationship between school rank and grad school acceptance, after accounting for GPA and GRE score?

Nested models

$$H_0 : \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A : \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Full model (H_A):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \\ \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Reduced model (H_0):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i$$

Comparing nested models

Full model (H_A):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i + \beta_3 \text{Rank2}_i + \beta_4 \text{Rank3}_i + \beta_5 \text{Rank4}_i$$

Reduced model (H_0):

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{GPA}_i + \beta_2 \text{GRE}_i$$

How do we compare these two nested models?

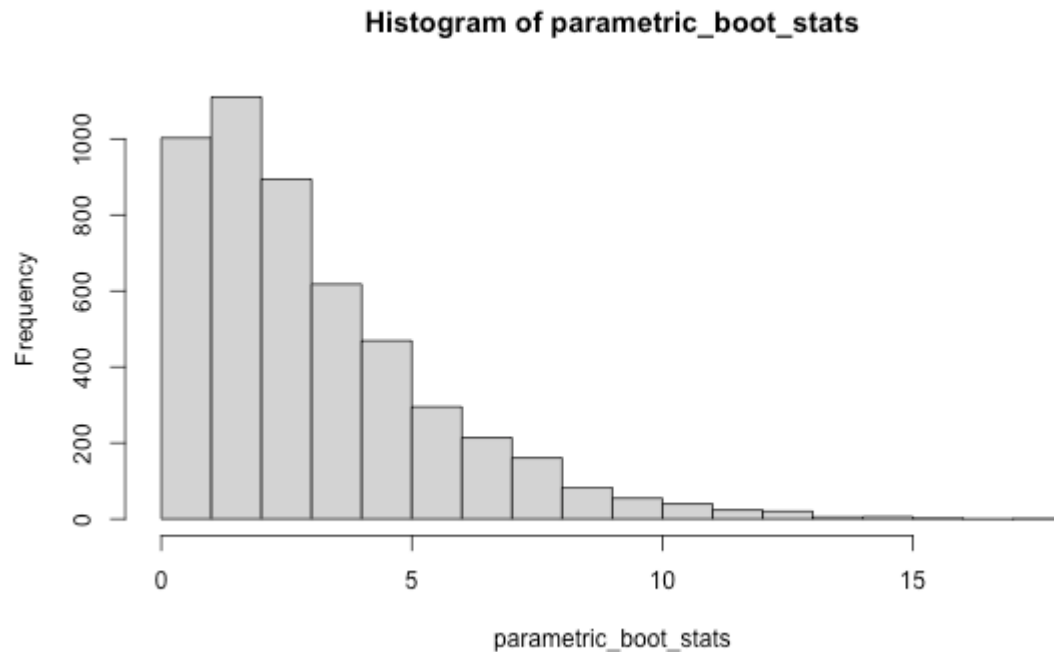
Comparing deviance

```
m_full <- glm(admit ~ gpa + gre + as.factor(rank),  
              family = binomial, data = grad_app)  
  
m_reduced <- glm(admit ~ gpa + gre,  
                 family = binomial, data = grad_app)  
  
m_reduced$deviance - m_full$deviance
```

```
## [1] 21.82649
```

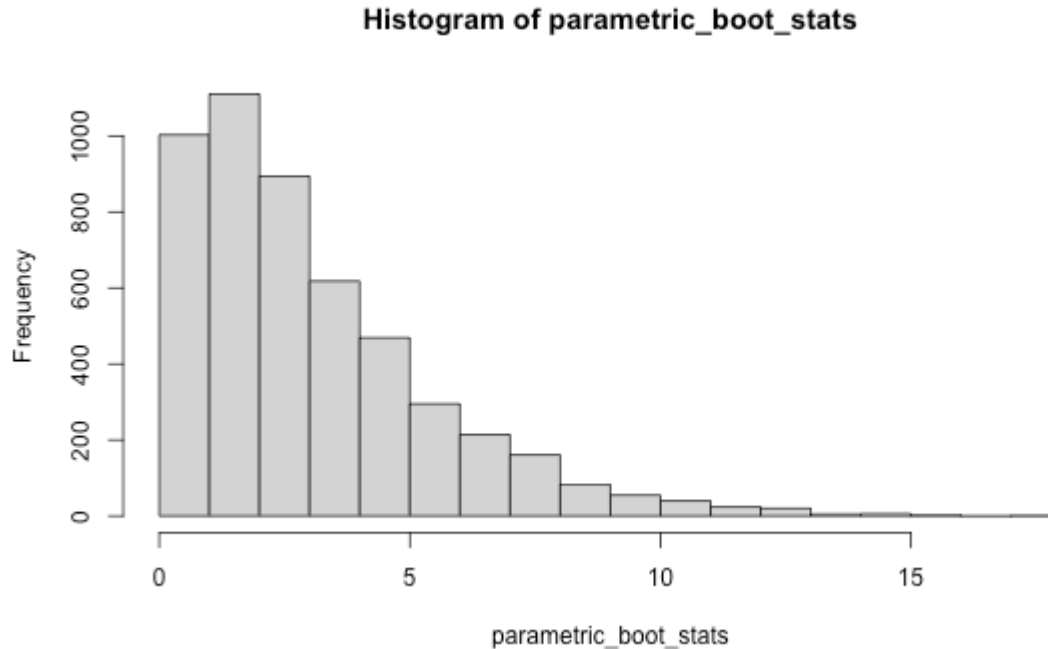
How do I know whether this is a big change in deviance? That is: is a change in deviance of **21.83** unusual under the null hypothesis?

Last time: Activity



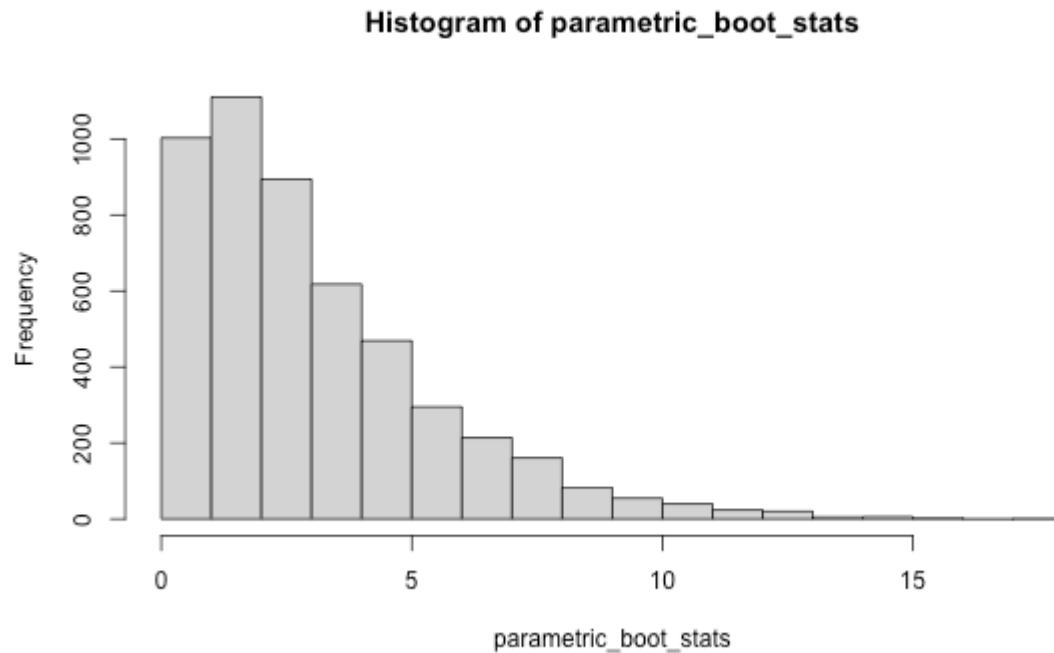
What does this distribution look like?

Last time: Activity



How unusual is the observed test statistic of 21.83, compared to this null distribution?

p-value

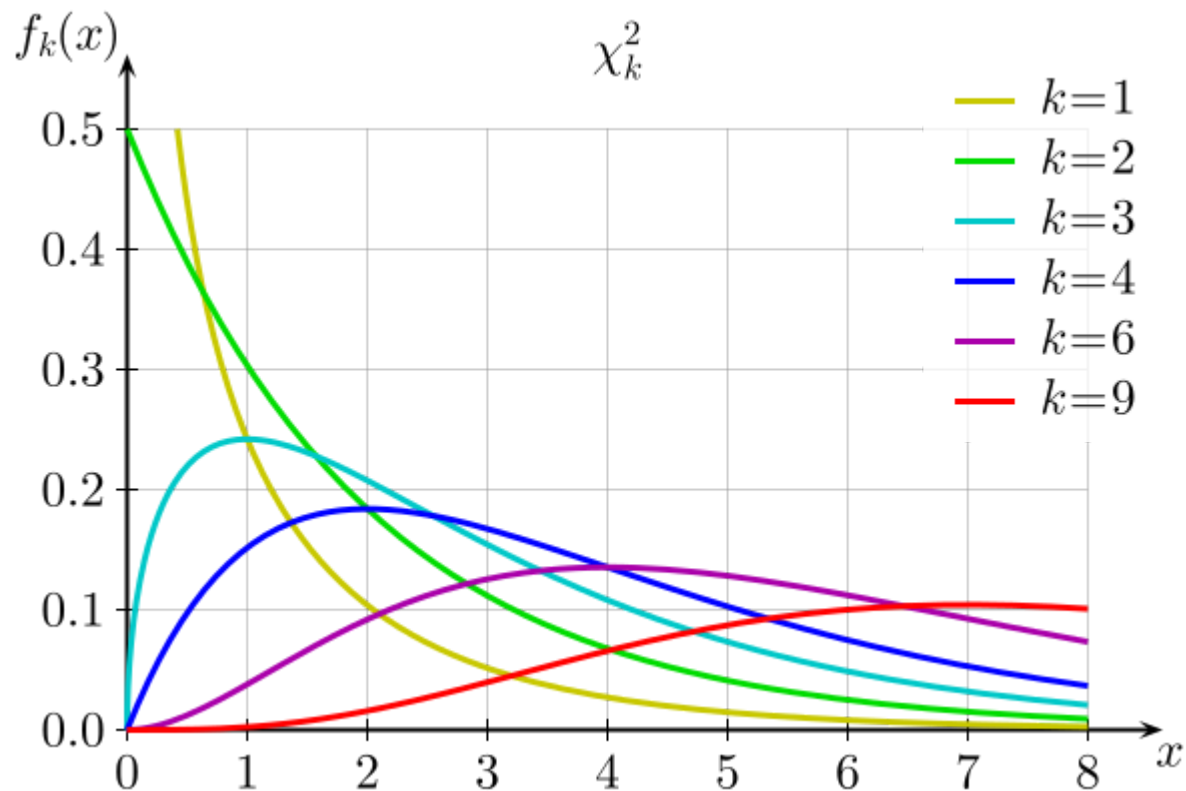


```
mean(parametric_boot_stats > 21.83)
```

```
## [1] 0
```

Null distribution

Under H_0 , $G \sim \chi_k^2$



Activity

- + Mean of χ_k^2 is k
- + Variance of χ_k^2 is $2k$

```
mean(parametric_boot_stats)
```

```
## [1] 3.054035
```

```
var(parametric_boot_stats)
```

```
## [1] 6.029801
```

Drop-in-deviance test (aka likelihood ratio test)

Goal: Compare nested models (full and reduced models)

Steps:

- + Calculate deviance for full and reduced models
- + Test statistic: $G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$
- + Null distribution: under H_0 , $G \sim \chi_q^2$

Activity 1

Work on the class activity (handout).

Activity

$$\text{Exercise}_i \sim \text{Bernoulli}(\pi_i)$$

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Loss}_i + \beta_2 \text{Smoke}_i + \beta_3 \text{HealthPlan}_i \\ + \beta_4 \text{Age}_i + \beta_5 \text{Height}_i$$

Research question: After accounting for the respondent's desired weight loss, is there any relationship between the other explanatory variables and exercise?

What are my null and alternative hypotheses?

Activity

Full model:

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1\text{Loss}_i + \beta_2\text{Smoke}_i + \beta_3\text{HealthPlan}_i \\ + \beta_4\text{Age}_i + \beta_5\text{Height}_i$$

Reduced model:

Activity

Full model:

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Loss}_i + \beta_2 \text{Smoke}_i + \beta_3 \text{HealthPlan}_i \\ + \beta_4 \text{Age}_i + \beta_5 \text{Height}_i$$

Reduced model:

$$\log\left(\frac{\pi_i}{1 - \pi_i}\right) = \beta_0 + \beta_1 \text{Loss}_i$$

Activity

Model 1:

Null deviance: 22680 on 19999 degrees of freedom
Residual deviance: 22036 on 19994 degrees of freedom

Model 2:

Null deviance: 22680 on 19999 degrees of freedom
Residual deviance: 22531 on 19998 degrees of freedom

Activity 2

Work on the class activity (handout).

Activity

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.0928484	0.2890071	-7.242	4.44e-13	***
loss	-1.4520863	0.1500420	-9.678	< 2e-16	***
smoke100	-0.1339504	0.0334117	-4.009	6.10e-05	***
hlthplan	0.5940486	0.0478039	12.427	< 2e-16	***
age	-0.0129321	0.0009882	-13.086	< 2e-16	***
height	0.0512637	0.0041823	12.257	< 2e-16	***

The standard deviation of the `loss` variable is 0.112.

How do the odds of exercise change with a one **standard deviation** increase in `loss` (not a one-unit increase!), holding the other variables constant?

Activity

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.0928484	0.2890071	-7.242	4.44e-13	***
loss	-1.4520863	0.1500420	-9.678	< 2e-16	***
...					

The standard deviation of the `loss` variable is 0.112.

Are we confident that a one-standard deviation increase in `loss` is associated with a decrease in the odds of at least 10% (holding other variables constant)?